1	2	3	4	5	6	Total
/20	/20	/20	/20	/20	/20	/100

Name: ______________________________ May 10, 2014 Saturday

Department/School:____________________ SID:__________________________

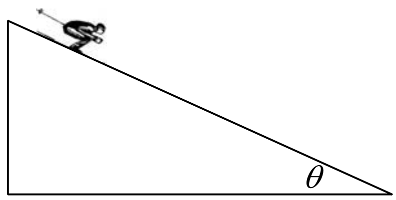
University Physics

Midterm Examination

Kuang Yaming Honors School, Nanjing University

<u>Select five out of following six problems.</u>

1. (20pts.) A skier of mass m moves down a track with constant slope θ. The motion is mostly subjected to the air resistance and the friction between the snowboard and the surface of snow is negligible. The air resistance is given by $f = -kv^2$. The initial velocity of the skier is zero.



(a) Write the equation of motion of the skier.

(b) What is the velocity of the skier as a function of time?

(c) What is the distance s traveled by the skier as a function of time?

(Hint: $\int \frac{dx}{a^2 - x^2} = \frac{1}{a}\tanh^{-1}\left(\frac{x}{a}\right) + C, \int \tanh(ax)dx = \frac{1}{a}\ln\cosh(ax) + C,$ $\sinh(x) = \frac{e^x - e^{-x}}{2}, \cosh(x) = \frac{e^x + e^{-x}}{2}, \tanh(x) = \frac{\sinh(x)}{\cosh(x)}.$)

2. (20pts.) A particle of mass m moves in one dimension along the x-axis. The force acted upon it is given by

$$F_x = \frac{\alpha}{x^3} - \frac{\beta}{x^2} \quad (x > 0)$$

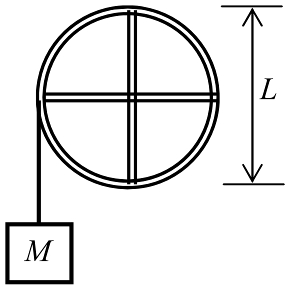
where α and β are positive constant.

(a) Find the potential energy function of the particle .(Choose the zero potential energy at infinity.)

(b) Find the position(s) of the equilibrium(s) of the particle, and identify it (them) as stable or unstable.

(c) If the particle is initially placed at the point $x = \alpha/\beta$, what is the minimum magnitude of velocity at that point the particle must have so that the particle can escape to the infinity.

3. (20pts.) A pair of long thin rods, each of mass M and length L, are connected to a hoop of mass M and radius $L/2$ to form a 4-spike wheel as shown in the figure. The center of the wheel is mounted to a fixed axel so that it can rotate freely in a vertical plane around the axle. Several turns of light string is wrapped on the wheel and a mass M is hanged on the one end of the string.

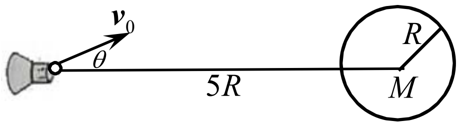


(a) What is the rotational inertia of the 4-spike wheel for the axis through the center of the wheel?

(b) Determine the tension T in the string.

(c) Calculate the angular acceleration α of the wheel and the linear acceleration a of the mass M.

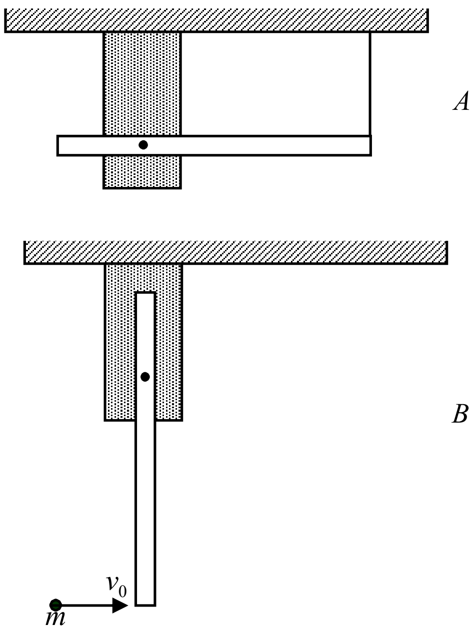
4. (20pts.) A spaceship is sent to investigate a planet of mass M and radius R. The ship first stays motionlessly in the position which is at a distance $5R$ from the center of the planet and then launches an instrument package with speed v_0. The mass of the package m is much smaller than the mass of the spaceship. The package is launched at an angle θ with respect to the straight line that links center of the planet and the spacecraft, as shown in the accompanying figure.



(a) For what angle θ will the package just graze the surface of the planet? What is the minimum initial speed v_0 the package required so that it is possible to graze the surface of the planet?

(b) What is the speed of the package when it grazes the surface of the planet?

5. (20 pts.) A long thin rod of mass M and length L is free to pivot around a fixed pin which is located at $L/4$ from the left end of the rod. The rod is held in a horizontal position by a string hanging vertically from the ceiling attached on the far right end of it, as shown in the figure A.



(a) Find the rotational inertia of the rod relative to the pivot. Calculate the tension T in the string that supports the rod.

(b) Once the string is cut, the rod begins to rotate around the pivot. Find the angular acceleration of the rod at the moment the string is just cut. Determine the angular velocity of the rod at the moment it reaches the vertically-oriented position.

(c) The rod is at rest in its vertically-oriented position and a ball of mass $m = 7M/12$ moving horizontally to the right with speed v_0 strikes elastically at the lower end of the rod, as shown in the figure B. What is the velocity of the ball after the collision?

6. (20pts.) To explain the puzzle of the "missing mass" of the universe, physicists hypothesize the existence of "dark matter", a cloud of unknown particles surrounding the centers of the galaxies. Assume a spherical galaxy is made (almost entirely) of the "dark matter", with uniform mass density ρ.

(a) Find the gravitational force acted on a star of mass m at a distance r from the center of the galaxy. (Assume that the dimension of the galaxy is larger than r.)

(b) If we use the galactic center as the origin of the coordinates, and assume the star moves purely in the radial direction, show that the motion is a simple harmonic one with a period $T = \sqrt{3\pi/G\rho}$.

(c) Suppose the star is at rest at $r = r_0$ when $t = 0$. Find the position and the velocity of the star as the functions of the time t.